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Interoperability and energy optimization in low pressure ship to medium pressure terminal interfacing

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ABSTRACT

Large-scale deployment of Carbon Capture and Storage (CCS) is essential to meet global climate change mitigation goals, with CO_2 shipping playing an important role in enabling cost-effective transport from dispersed sources to storage sites. However, transporting CO_2 is facing interoperability challenges due to the absence of standardized specifications for transport conditions, including state (gas, liquid, or dense phase), pressure and temperature levels.

To address this, the authors propose a compatibility framework between Low Pressure (LP, 7–10 bara) ships and Medium Pressure (MP, 12–20 bara) terminals, serving liquefied CO_2 (LCO_2) shipping. The aim is to identify machinery systems and operating strategies that ensure cost-effective LCO_2 transfer. A thermodynamic model is used to simulate and optimize performance during loading and discharging operations, by quantifying energy and equipment needs and proposing ship-terminal interface solutions. The model is applied in a case study featuring a ship with three 7300 m^3 tanks (simulated range: 5840–8760 m^3) and evaluated across varying pressures, temperatures, and flow rates. The study concludes that:

- During LP discharging, the LCO_2 is pressurized and heated to MP saturation conditions. To reduce vapor displacement from the terminal tank, subcooled inlet conditions are applied. Vapor returned to the ship is cooled via heat integration before being depressurized to LP.
- During MP loading, the fluid stream is depressurized to LP, requiring subcooling to avoid two-phase flow and excessive vapor return. A sub-cooler reduces vapor displacement but demands more energy than a high-capacity vapor return compressor.

1. Background

1.1. Challenges of the CO_2 shipping-terminal compatibility

The large-scale deployment of Carbon Capture and Storage (CCS) is essential for achieving climate neutrality, with shipping transportation emerging as an important enabler to connect CO_2 capture sources to sequestration sites. However, a growing challenge in the CCS value chain lies in the lack of standardized CO_2 conditions between projects, including the pressure and temperature levels of liquefied CO_2 (LCO_2) shipping and terminal infrastructure (Zhang et al., 2024).

CO_2 shipping with optimized liquefaction and terminal design is a cost-effective alternative to pipelines for large-scale CCS transport over medium to long distances, Yoo et al. (2013), Deng (2023), Deng et al. (2019). Large-scale transportation favours low-pressure (LP) regimes (e.g., 7–10 bara), due to the higher density, which allows for higher capacities for mass transport at the same volume, CETO (2025), Cha, D. (2024). In contrast, current industrial standards and flagship CCS project specifications across the EU adopt medium-pressure (MP) regimes (12–20 bara) for CO_2 terminals, as demonstrated by various projects.

CCS projects vary significantly in ship and terminal designs, particularly in intended operating pressures and temperatures, creating challenges for interoperability across networks. To date, most operational experience in CO_2 shipping has been with systems designed for MP conditions, as seen in existing food-grade LCO_2 carriers, which operate at smaller capacities aligned with industrial supply chains. Northern Lights (Norway) currently employs vessels of around 7500 m^3 capacity

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Nomenclature

BOG	Boil-off Gas
BOR	Boil-off Rate
CAPEX	Capital expenditure
CCS	CO2 Capture and Storage
COP	Coefficient of Performance
COSSMOS	Complex Ship Systems Modelling and Simulation
HTX	Heat exchanger
IPCC	Intergovernmental Panel for Carbon
JT	Joule-Thomson
LCO_2	Liquefied Carbon Dioxide
LMTD	Logarithmic Mean Temperature Difference
LP	Low pressure
MARV	Maximum Allowed Reference Value
MP	Medium pressure
SRK	Soave-Redlich-Kwong (SRK). equation of state
VR	Vapor return
ΔT	Temperature differential

Symbols

A	Heat exchanger surface (in m^2)
h	Mass specific enthalpy (in J/kg)
$LMTD$	Logarithmic mean temperature differential (in °C)
$\dot{m}$	Mass flowrate (in kg/s)
T	Temperature (in °C)
Q	Thermal duty power (in W)
U	Heat transfer coefficient (in $W/(m^2 \times °C)$)
LIQ	Subscript: Liquid
TER	Subscript: Terminal
VR	Subscript: Vapour return

for its initial phase, Northern Lights (2025).

In contrast, emerging LP concepts are driving a shift in the LCO_2 carrier newbuilding orderbook, aiming to enable larger capacities and improved transport economics to meet the growing demand for high-volume CCS transport. DNV, in collaboration with Shell, Equinor, TotalEnergies, and Gassco, is working on a Joint Industry Project called CETO (CO_2 Efficient Transport via Ocean), which focuses on the technical qualification of a 30,000 cbm low-pressure LCO_2 carrier design, CETO (2025). The project has delivered a detailed ship and cargo tank concept addressing key challenges of large-scale LP LCO_2 transport, including material performance, fatigue resistance, and operational safety.

At this stage, the exact specifications for emerging European CO_2 terminals remain undefined, highlighting the need for adaptable facilities, which offer open-access to variant regimes, in alignment with the CCS Directive Article 21. These terminals aim to be developed as open-access facilities, capable of accommodating vessels with varying design parameters and operating conditions. CO_2next terminal (Netherlands), part of the Aramis initiative, is designed for temporary storage and conditioning of CO_2 to ensure compatibility with offshore pipeline and storage platforms, CO2next (2025). Greensand terminal (Denmark) includes a transit facility supporting CO_2 handling for injection at the Nini West field, Greensand (2025). Ghent Carbon Hub (Belgium), developed by Fluxys, ArcelorMittal, and North Sea Port, will liquefy and export CO_2 from local industry, Ghent Carbon Hub (2025). Dunkirk CO_2 export terminal (France), also developed by Fluxys and integrated with the Dunkerque LNG site, is intended to serve as a major hub for CO_2 captured in northern France, enabling large-scale export and offshore storage, D'Artagnan Dunkirk CO2 Hub (2025). In November 2025, the APOLLOCO$_2$ project secured €169.3 million EU funding to advance large-scale CO_2 transport and storage in South-East Europe. Led by DESFA and Ecolog, it will develop a CO_2 hub in Revithoussa island integrating LNG regasification with CO_2 liquefaction to cut power use and costs, and enabling flexible shipping. Recognized by the EU as a leading CCS innovation, the project connects emitters and storage via pipelines and shipping.

As the European CO_2 value chain evolves, it is likely to develop as a patchwork of projects with differing technical specifications and operational regimes. This fragmentation will necessitate solutions that enable interoperability and open-access terminal operations, despite variations in pressure regimes.

1.2. The need of compatibility assessment

The lack of standardization between shipping and terminal specifications creates technical and economic incompatibilities at the ship-terminal interface. LP ships, optimized for large-scale transport, may face operational constraints or require costly adaptations when interfacing with MP terminals. Without harmonized specifications or interfacing solutions, the lack of standardization in CO_2 transport conditions risks undermining the efficiency, scalability, and interoperability of emerging CCS networks.

2. Scope of work

2.1. Objective and methodology

This study aims to investigate the operational interface between a LP CO_2 ship and a MP terminal, focusing on the efficient and safe transfer of LCO_2 across differing pressure regimes. A stepwise approach is followed:

- Step 1: First, a market review is conducted to identify viable technologies and components that could define the ship-terminal interface.
- Step 2: A model-based analysis is carried out using simulation tools to evaluate the machinery and systems required for pressure-level interfacing between the ship's and terminal's CO_2 tanks. These simulations provide critical insights into the technical challenges and operational requirements of the transfer process. Operational constraints are also modelled, with a focus on completing loading and discharging cycles within a 12-hour duration. This includes evaluating heating and cooling needs to maintain process efficiency and safety under varying conditions.
- Step 3: A feasibility assessment is then performed to evaluate the proposed solutions against current state-of-the-art technologies. Additionally, a sensitivity analysis is conducted to assess the impact of varying tank capacities and pressure levels. These results are visualized using in p-H diagrams to illustrate different operational scenarios.
- Step 4: Finally, the study determines the optimal sizing of key components, machinery, and systems, offering recommendations based on the operational ranges of available equipment both onboard and at the terminal. Energy consumption is quantified in kilowatt-hours per unit mass of CO_2, ensuring that the proposed solutions are not only technically viable but also energy-efficient and sustainable.

For the process simulation, DNV's COSSMOS software (Complex Ship Systems Modelling and Simulation) was used, Dimopoulos et al. (2014). COSSMOS specializes in modelling, simulation, and optimization of ship machinery systems, covering both traditional and new technologies. The tool supports the virtual development of engine

rooms, to facilitate energy and environmental assessments, and enable the optimization of marine energy systems.

2.2. Case study specifications

The case study specifications are summarized in Table 1 and Fig. 1. The vessel is a class 22,000m^3 LCO_2 carrier, with main particulars as shown in Table 1, while the terminal tank is considered at an intermediate storage capacity of 30,000 m^3. The analysis is conducted over a range of ship and terminal pressures and containment capacities, such that the transported mass remains constant at the reference value of approximately 24,700tons of LCO_2. The range of pressure regime definition is given in Fig. 1. The loading and discharging rates are estimated through simulation, targeting a total operation time of approximately 12 h under varying tank volumes, pressure conditions and liquid line preconditioning. The LCO_2 composition is according to Northern Lights Joint Venture (2024).

2.3. Discharging process specifications

Fig. 2 presents a graphical illustration of the overall discharge system. The interface system between ship and terminal tanks must ensure compatibility between the ship's LP and the terminal's MP conditions. The system includes two main transfer lines: (a) the LCO_2 discharging line and (b) the vapor return (VR) line, which is necessary to maintain ship tank pressure. Onboard cargo pumps pressurize the LCO_2 to match the terminal tank pressure, with additional head considered for MP conditions, while onshore heat exchangers (HTX) warm the LCO_2 to terminal saturation temperature before storage. At the same time, onshore coolers condition the vapor returning to the ship to match tank conditions. Onboard VR control valves regulate flow based on the ship's cargo tank pressure set point. At the terminal tank side, vapor generation during loading must be managed to prevent over-pressurization and ensure stable operation.

During the discharging operation, LCO_2 is transferred from the ship's cargo tanks to the terminal storage tank via the liquid header, ship's manifolds, and marine loading arms. Prior to discharging, the ship tanks are filled to 96 % of their volume with LCO_2. The operation concludes when the tanks retain a minimum heel for the ballast voyage, e.g., 4 % volume. The process includes five (5) thermodynamic transitions, as depicted in the phase diagram of pure CO_2, Fig. 3:

Table 1
Case specifications.

Position	Feature	Value	Unit
Ship	Number of ship cargo tanks	3	-
	Pressure of ship cargo tanks	7 - 10	bara
	Discharging duration	12	Hr
	Filling level	96 %	%
	Tank pressure	7; 8.5; 10	bara
	Tank volume	5840; 7300; 8760	m^3
	Tank temperatures	−49; −44; −40	°C
	LCO_2 density (liquid)	1146; 1123; 1101	kg/m^3
	CO2 density (vapor)	18; 22; 26 kg/m^3	kg/m^3
	Loading rate (at MP 16, LP 7 bara, 7300 m^3)	1724	m^3/hr
	Discharging rate (at MP 16, LP 7 bara, 7300 m^3)	1679	m^3/hr
Terminal	Tank pressure	12; 16; 20 bara	bara
	Tank volume	30,000 m^3	m^3
	Tank temperatures	−35; −27; −20 °C[3]	°C
	LCO_2 density (liquid)	1079; 1035; 989	kg/m^3
	CO2 density (vapor)	31; 41; 52	kg/m^3

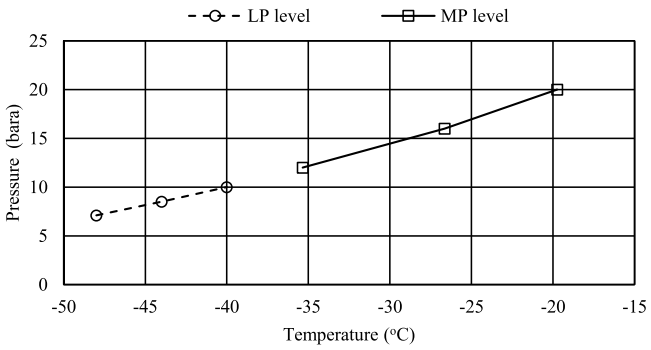


Fig. 1. LCO_2 pressure and temperature thresholds examined in this study.

- A-B: LCO_2 is pressurized using centrifugal pumps to match the terminal's MP tank level. This can be achieved either with onboard deep-well cargo pumps, provided they offer sufficient head, or with the addition of booster pumps. The system must overcome the pressure differential between the LP ship and MP terminal, as well as line pressure losses. The modelled discharging rate is set to complete a 7,300m^3 tank transfer within 12 h.
- B-C: An onshore HTX is used to raise the temperature of the LCO_2 to match the terminal tank's saturation conditions. The shore side cooler is assumed to be a seawater-driven one.
- C-D: As the terminal tank fills, vapor is displaced by the incoming liquid due to the piston effect. A portion of this vapor is returned to the ship to maintain tank pressure, while the excess must be treated. If the incoming liquid is subcooled, vapor condensation occurs in the terminal tank, reducing the volume of excess vapor.
- D-E: The vapor returning to the ship is cooled to reach saturated vapor conditions, achieving a lower temperature by the end of the process.
- E-F: Finally, the VR stream is depressurized to match the ship's LP tank conditions, completing the vapor handling cycle.

2.4. Loading process specifications

During the loading of CO_2 from a terminal to a LP ship, the control of temperature and pressure is essential to ensure safe and efficient transfer. The interface system, Fig. 4, includes components for conditioning both the liquid stream and the VR. At the side of the loading line, onshore HTXs are used to cool the liquid CO_2 stream, especially when subcooling is required to minimize vapor generation in the ship tank. Onboard Joule-Thomson (JT) valves depressurize the liquid to match the ship tank pressure, enabling controlled entry into the cargo tanks. At the side of the VR line, onshore VR compressors increase the pressure of the returning vapor to align with terminal conditions. Moreover, onshore HTXs cool the vapor stream to the terminal's required temperature. Pressure-controlled liquid cargo valves (located either onboard or onshore) regulate flow and maintain system stability during the transfer.

During the loading operation, LCO_2 is transferred from the terminal tank to the ship's cargo tanks. Upon arrival at the export terminal, the ship tanks contain the minimum heel, which helps maintain pressure during the ballast voyage. The loading process concludes when the tanks reach 96 % of their volume with LCO_2. The key steps of the loading process are outlined below and illustrated in the simplified process diagram of Fig. 5:

- A-B: The LCO_2 stream transitions from MP terminal conditions to LP ship conditions. This pressure drop can generate significant vapor. To mitigate this, a sub-cooler (A-B) is used to reduce the temperature of the liquid before depressurization, thereby reducing vapor

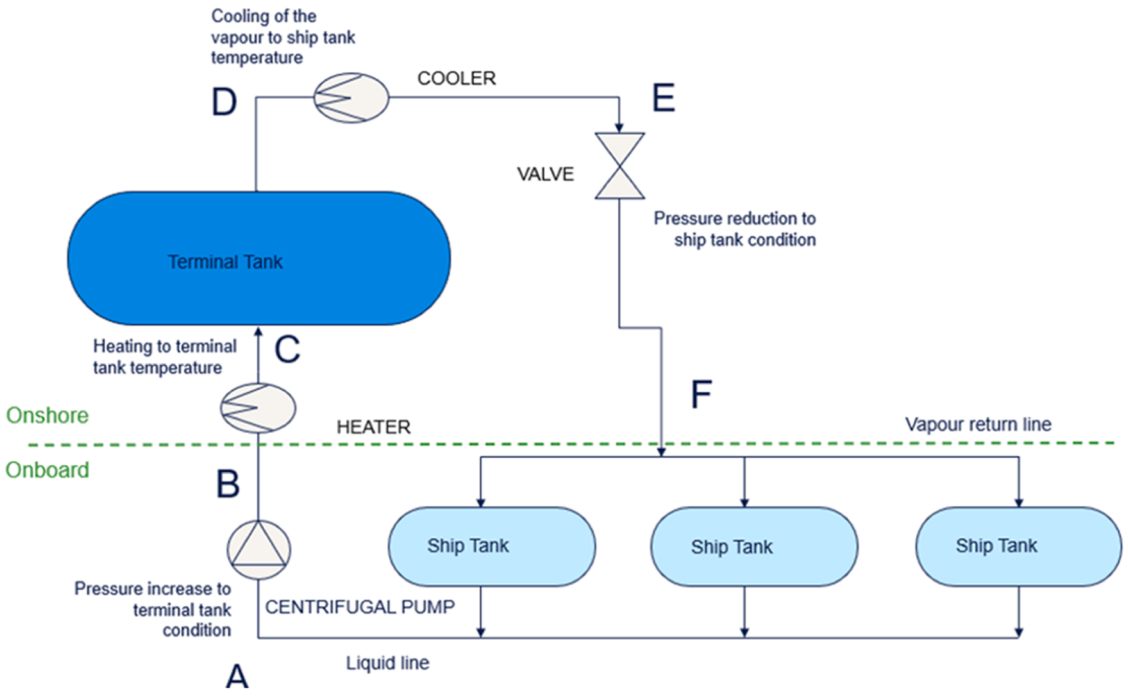


Fig. 2. Discharging process flow diagram. The letters denote key positions of thermodynamic transitions.

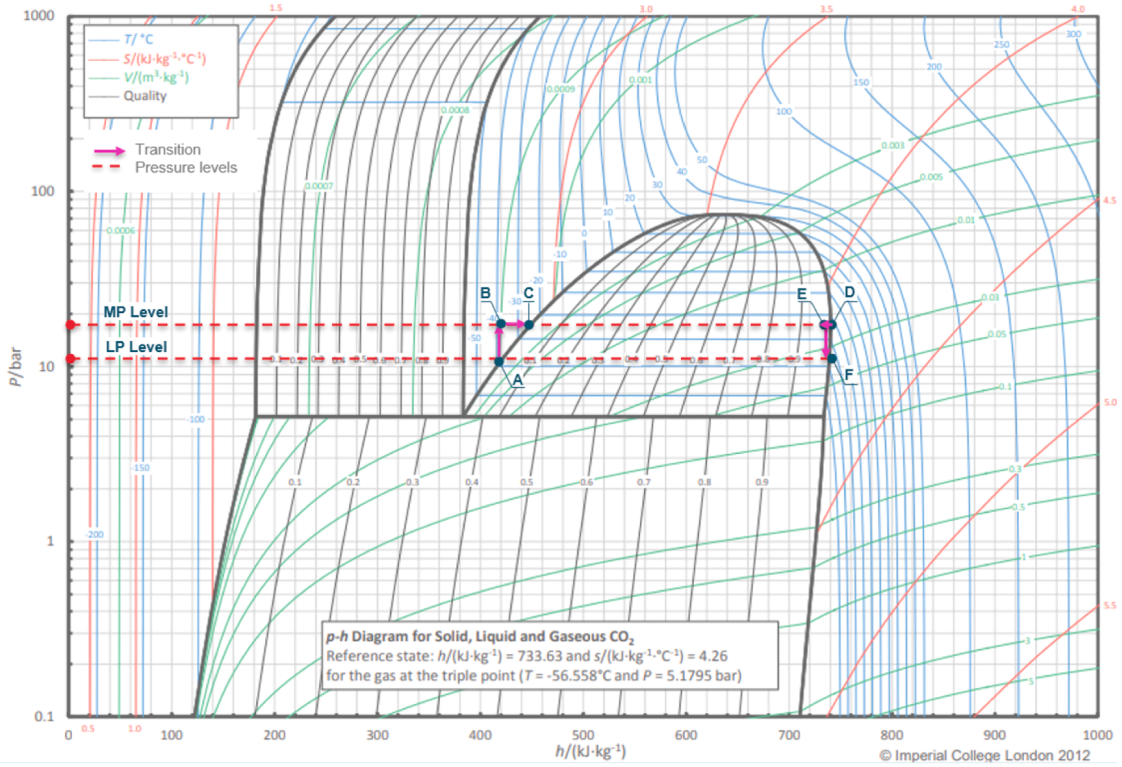


Fig. 3. Illustration of the thermodynamic transitions of LCO_2 in the pressure-enthalpy diagram of pure CO2, during discharging. The phase diagram used is found in (Imperial College London, Thermophysics Research Group n.d).

generation and the volume of displaced vapor. The pressure reduction is achieved via a JT valve.

- C: Vapor generation within the ship tanks must be carefully managed to maintain pressure within the ship's operational envelope. This ensures safe and stable loading conditions.
- D-E: Vapor displaced from the ship tanks is returned to the terminal via a VR line. Dedicated reciprocating VR compressors are used to raise the vapor pressure to terminal tank levels.
- E-F: After compression, the "hot" vapor is cooled down using a HTX to match the terminal tank conditions, completing the vapor handling cycle.

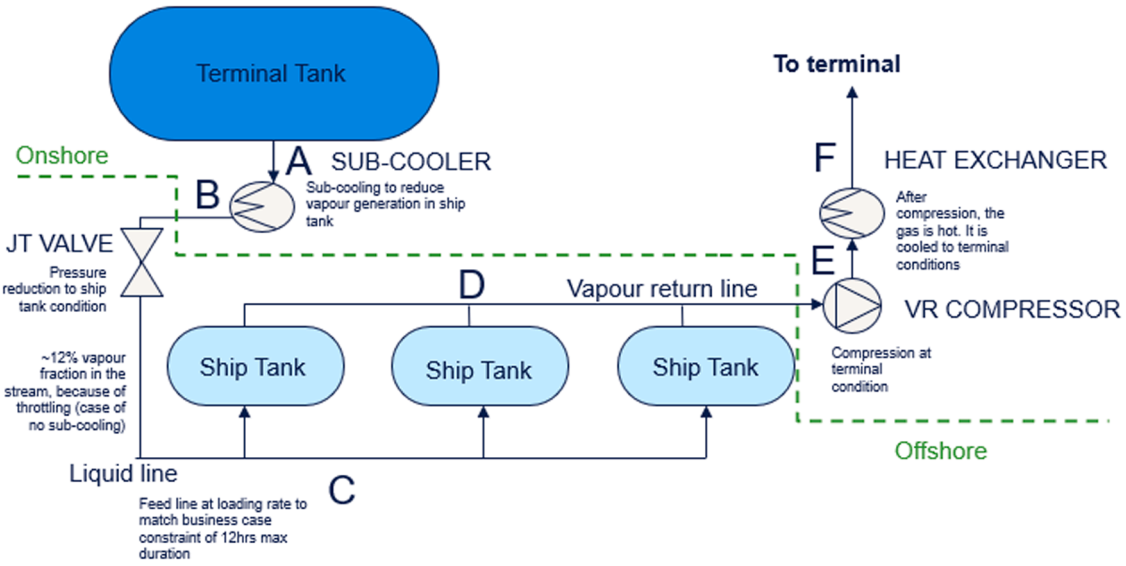


Fig. 4. Loading process flow diagram. The letters denote key positions of thermodynamic transitions.

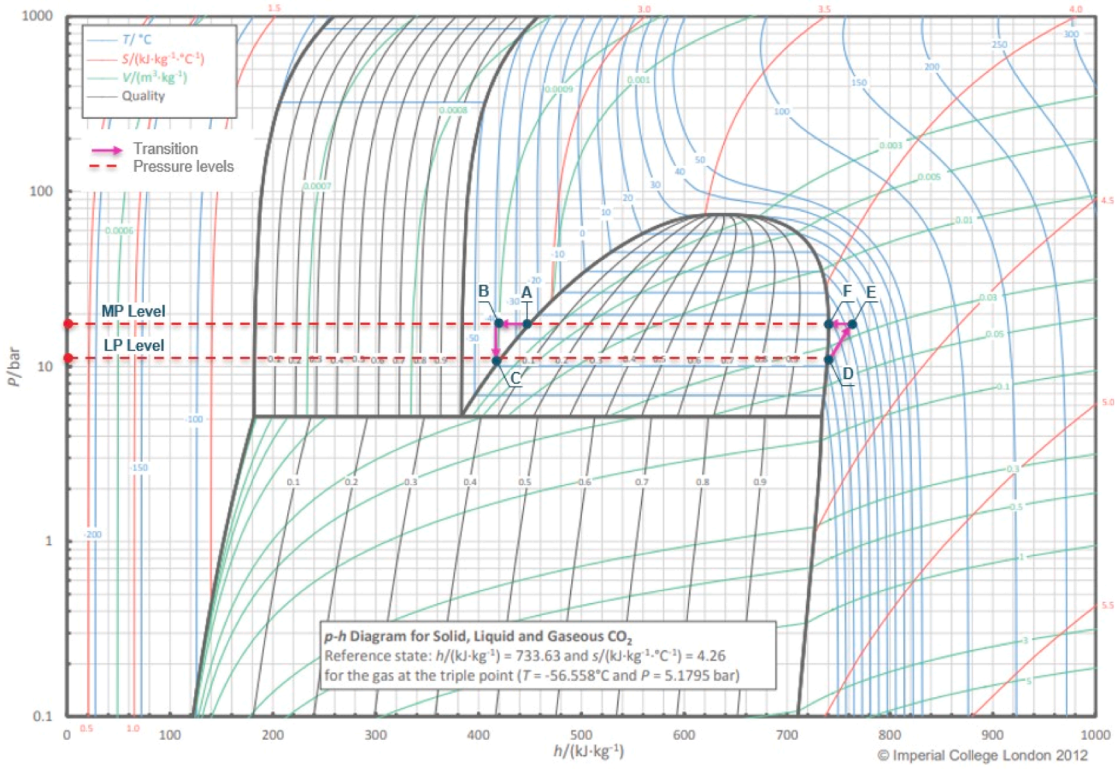


Fig. 5. Illustration of the thermodynamic transitions of LCO_2 in the pressure-enthalpy diagram of pure CO2, during loading. The phase diagram used is found in (Imperial College London, Thermophysics Research Group n.d).

2.5. Performance parameters and assumptions

The discharging process is governed by a series of key sub-tasks; each monitored through specific performance parameters. The performance of the LCO_2 line conditioning is monitored via (a) the pump head and required duty power and (b) the heating duty, to pressurize and heat the liquid line from LP to MP, respectively. The ship tank pressure management performance is monitored through the VR amount needed for the overall process and the associated cooling requirements. The terminal tank pressure control is assessed via the total displaced and excess vapor volume.

The vapor generation (in tons) at the ship tank during the loading process is an important performance parameter, as it affects the timing of the process. As described in paragraph 2.4, the loading process can be optimized thermodynamically through the subcooling of LCO_2 prior to its entry into the ship tanks, with the goal of minimizing vapor generation. Depending on the vapor generated, the following parameters are quantified: (a) the power demand for vapor compression; (b) the power demand for temperature conditioning of returned vapor; and (c) the sizing of the respective compressors and coolers.

In both the discharging and loading processes, the specific demands are calculated. The metric is defined in power in kilowatt-hours (kWh) per ton of LCO_2 transferred. Furthermore, all cases are analysed following the below assumptions:

- CO_2 purity specifications according to Northern Lights project specifications, Northern Lights (2025).
- The ship and terminal tanks are modelled as shown in Dimopoulos et al. (2025). The model captures vapor liquid equilibrium at the liquid bulk and superheated conditions at gas phase. The cubic equation of state of Soave-Redlich-Kwong (SRK) is used to calculate CO2 mixture thermophysical properties, Soave, G. (1972), Diamantonis et al. (2013).
- The Logarithmic Mean Temperature Differential method is considered for HTX sizing, combined with empirical values for the heat transport coefficient, Perry et al. (1999).
- The centrifugal pump efficiency is assumed to be fixed to 75 %.
- The VR compressor is modelled using polytropic thermodynamic transition, and assuming a nominal isentropic and electric efficiencies: 80 % and 96 %, respectively.

3. LP-MP discharging operation

3.1. Evaluation of the LP to MP discharging requirements

Before its entry to the terminal tank, the LCO_2 is heated to reach the terminal temperature level. Through the discharging process simulation, the heating demand of the required HTX is estimated, for different pressure values of low and MP, as shown in Fig. 6(a). The heater operates with seawater as heating medium. The seawater pump demands to provide the heat load of Fig. 6(a) is estimated at a low range of 325 to 395 kW. A pump is required to pressurize LCO_2 from LP to MP pressure condition. The available deep-well cargo pump specifications (inside cargo tanks) shall cover the above requirements, as shown in Fig. 6(b). The estimated range of required pump specifications and the power demand per cargo ton is presented in Table 2.

VR controls the ship tank pressure at preferred level within the operational envelop. The vapor flows from terminal to ship, through the VR line, under the pressure difference imposed by the piston effect acting on the vapor by the rising liquid interface in the terminal tank. It must be noted that due to density difference, the vapor mass in the terminal tank to be displaced is more than the one required by the ship at ballast voyage.

To analyse this process, the discharging operation was simulated considering three different pressure values for the ship tanks and three different pressure values for the terminal tanks (see Table 1 for values). As a result of the simulation, it was observed that changes to MP pressure did not impact the LP vapor demand. With the rise in ship tank pressure, both the quantity of required vapor and the vapour density increase. The cooling of the return vapor (MP to LP temperature) was also calculated. The simulation results can be seen in Fig. 7, which present the VR flow and the required cooling duty.

During the transfer from the ship to the terminal tank, vapor is generated at the terminal tank (Fig. 8). A portion of this vapor is returned to the ship's tank to maintain pressure (Fig. 7(a)), while the remainder is managed by the terminal. Fig. 8 presents the excess vapor estimation for a range of ship tank pressures (7; 8.5; 10 bara).

Table 2
Pumps specifications.

Feature	Value	Unit
Pump power demand (per ship tank)	100–750	kW
Pump capacity (per ship tank)	up to 600	m^3/h
Pump Head	140	meters
Pressure	up to 20	bara

3.2. Improving the LP to MP discharging process

To support the optimization of the LP-MP discharging process, several potential approaches are being explored. This paragraph presents an evaluation of these strategies, acknowledging that their effectiveness may vary depending on specific operational conditions and system configurations.

- **Option 1 – LP to MP line conditioning at sub-cooled conditions:** One approach involves adjusting the liquid line temperature (sub-cooling) to potentially minimize vapor formation at the terminal side; this would require careful evaluation of how liquid line inlet temperature influences vapor generation in terminal tanks.
- **Option 2 – Integration of VR with the liquid line:** Another concept is the thermal integration of the VR line with the liquid line, using the latter to cool the returning vapor, which may help reduce vapor accumulation in the terminal tank under favourable conditions.
- **Option 3 – Spray cooling:** The ship spray cooling system could be utilized to lower the available vapor mass and reduce the vapor cooler duty, though the optimal impact would depend on system dynamics and cargo properties.

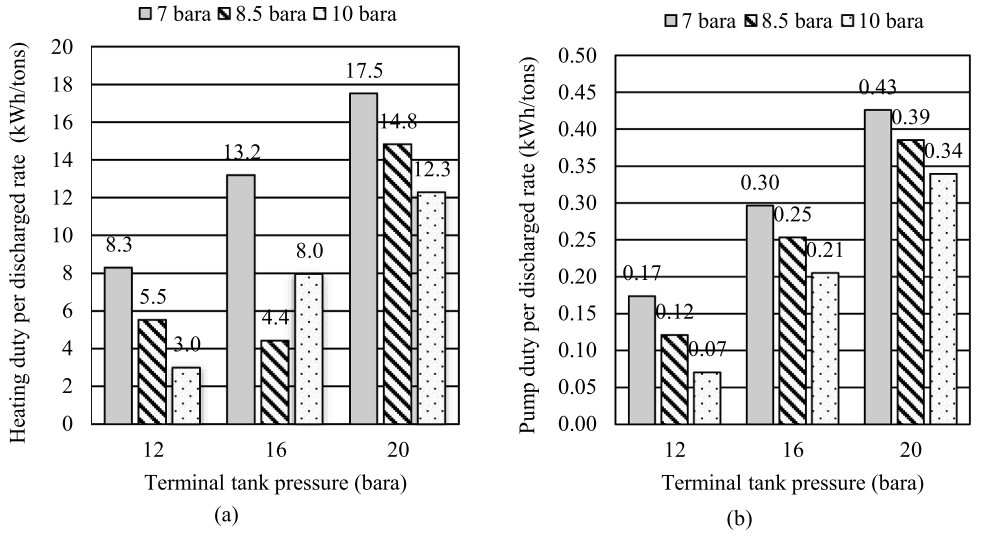


Fig. 6. Demands for conditioning the liquid discharging line: (a) Thermal duty and (b) pump duty per discharge rate as function of the ship tank pressure.

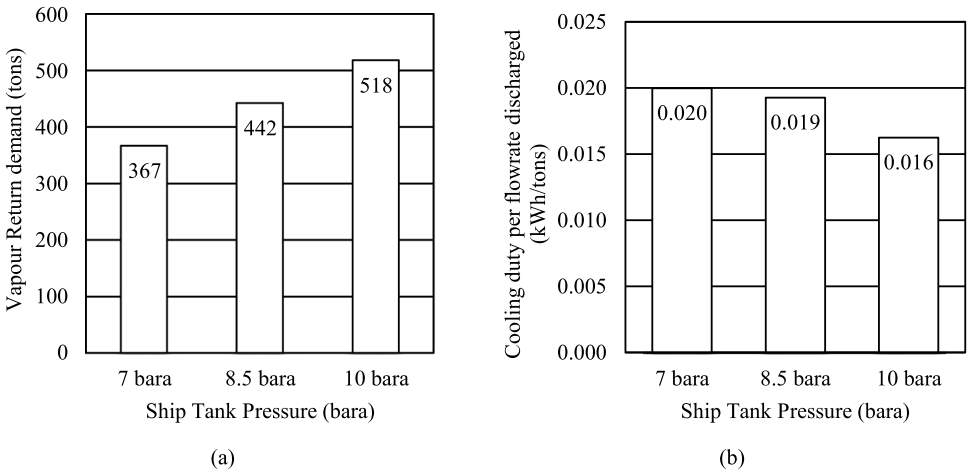


Fig. 7. Total VR demand in tons for the complete discharging process (a), and cooling duty to condition the VR to the ship at low temperature threshold (b). In all cases, the terminal tank MP is 16 bar.

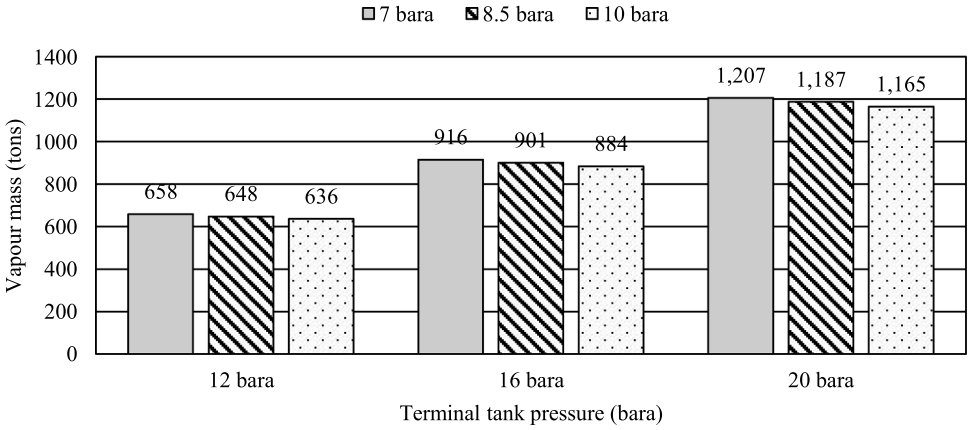


Fig. 8. Total vapor displaced in the terminal tank.

- **Option 4 – Integration:** Finally, the onboard vaporizer might be employed to preheat the liquid cargo, potentially easing the demand on the main heater and improving energy efficiency. These measures are subject to further validation and should be assessed within the specific operational and technical context of each vessel.

All the above approaches are analysed as follows.

3.2.1. Effect of subcooling in reducing terminal vapor displacement and liquid line conditioning demands

Simulation results suggest that vapor displacement in the terminal tank can be significantly reduced when the incoming liquid stream

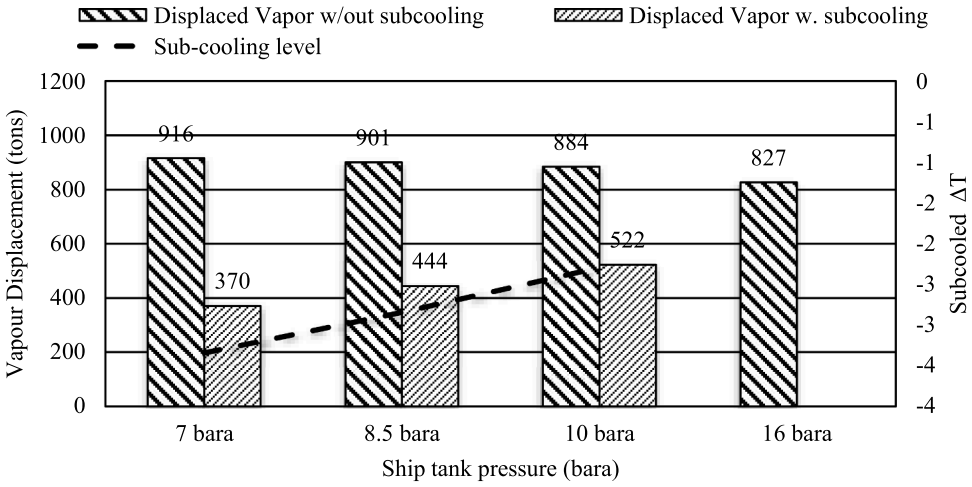


Fig. 9. Effect of subcooling on terminal tank vapor displacement. MP at 16 bara is considered.

remains subcooled -instead of being heated to match the saturated MP conditions of the terminal tank. This subcooling level can be fine-tuned to align with the VR requirements of the ship tank. Consequently, the heater can be designed to deliver only the necessary thermal duty. As illustrated in Fig. 9, the vapor mass in the terminal tank is notably lower when subcooling is applied. The subcooling level is plotted on the y-axis relative to the MP saturation temperature at 16 bara (−27 °C).

Subcooling influences the thermal requirements of the discharged liquid line, reducing both the nominal heat duty on the HTX and the demand for heating medium. This leads to lower capital expenditure (CAPEX) and reduced consumption for seawater circulation. Assuming a two-stage heating network, first operated with a glycol-based medium (LCO_2-glycol) and, second, by seawater, the required seawater flow rate was calculated based on a temperature differential (ΔT) of 7 °C. Alternatively, specialized heater equipment that protects from seawater freezing can be considered. The HTX thermal duty was also reassessed to account for the effects of subcooling. As shown in Fig. 10, subcooling results in a reduction in both heat duty and seawater flow rate, supporting a more sustainable and energy-efficient operation.

3.2.2. *Effect of integration of the vapor return with the liquid line to reduce demands for vapor return conditioning*

The integration of vapor and liquid line is assessed (option 2 of paragraph 3.2, illustrated in Fig. 11), considering a secondary network that supplies cooling from the liquid line to the vapor line. Such a system would be expected to be located at shore side. The cooling medium would be selected based on the desired temperature differentials and could be typical refrigerants for processes, like ethylene, propane or water-glycol, or other.

Simulating option 2, the liquid stream mass flowrate, which is required to perform conditioning of the vapor line, is <0.2 % of the total liquid mass flow, resulting in negligible benefit compared to the increased system complexity and higher CAPEX.

3.2.3. *Effect of spray cooling in reducing demands for vapor return conditioning*

The use of spray cooling to reduce demands for vapor cooling, by branching-off of the liquid line to ensure enthalpic mixing and temperature drop at vapor line, is explored, as shown in Fig. 12. As can be seen in the below diagram, an amount of LCO_2 from the ship tank can be recirculated and sprayed in the VR line, to cool down the vapor to ship tank conditions and minimize or even totally remove the need for the use of a cooler. The required mass flow rate of LCO_2 for spray cooling, to cover the full cooling demand, considering LP value equal to 7 bara and MP value equal to 16 bara, is calculated as per Eq. (1), based on the energy balance between the two phases:

$$\dot{m}_{LIQ} = \frac{\dot{m}_{VR} * (h_{VR} - h_{TER})}{h_{LIQ} - h_{TER}} \quad (1)$$

where h_{VR} is the enthalpy of the vapor in ship tank conditions (J/kg), $\dot{m}_{VR}$, is the mass of the vapor in ship tank conditions (tons/h), h_{LIQ} is the enthalpy of the LCO_2 after pumping (J/kg), $\dot{m}_{LIQ}$, is the required mass of the spray cooling (tons/h), h_{TER} is the enthalpy of the vapor in terminal conditions (J/kg), and $\dot{m}_{TER}$, is the mass of the vapor in terminal conditions (tons/h). Considering the already calculated VR mass for 7–16 bara case, the required mass of spray cooling is approximately 5.7 tons, which is 0.025 % of the total discharged mass from the ship tanks.

3.2.4. *Effect of vaporizer used to reduce liquid line conditioning demands*

The option of using an existing vaporizer onboard ship along with the pump and the heater before the transfer of the LCO_2 to the terminal tank was examined. The vaporizer can partially heat the liquid, thereby decreasing the power required from the HTX to bring the LCO_2 to the terminal temperature conditions. For the subject calculations, it was assumed that the ship's vaporizer has a capacity equal to the VR demand, as shown in Fig. 13, resulting in power demand within the range of 2–3.5 MW, for different combinations of LP-MP conditions.

In Fig. 14, the distribution of load between the vaporizer and HTX is illustrated for various LP-MP combinations. It is demonstrated that a small percentage of the total heating can be covered by the vaporizer, therefore, this option is not considered feasible.

3.3. *Discharge optimization results*

By consolidating the most effective integration practices identified through options 1,2,3 and 4, the HTX capacity and seawater flow rate are determined and the overall demand associated with the LP to MP discharge process is estimated, ensuring a comprehensive understanding of both thermal and operational performance under varying conditions.

First, the thermal duty (Eq.2) of the liquid conditioning is estimated. The thermal duty can be used as input to estimate the HTX design characteristics, based on the expected logarithmic temperature differential (Eq.3) and heat transport coefficient. The design characteristics were estimated for the different LP-MP combinations of Table 1, as shown in Table 3.

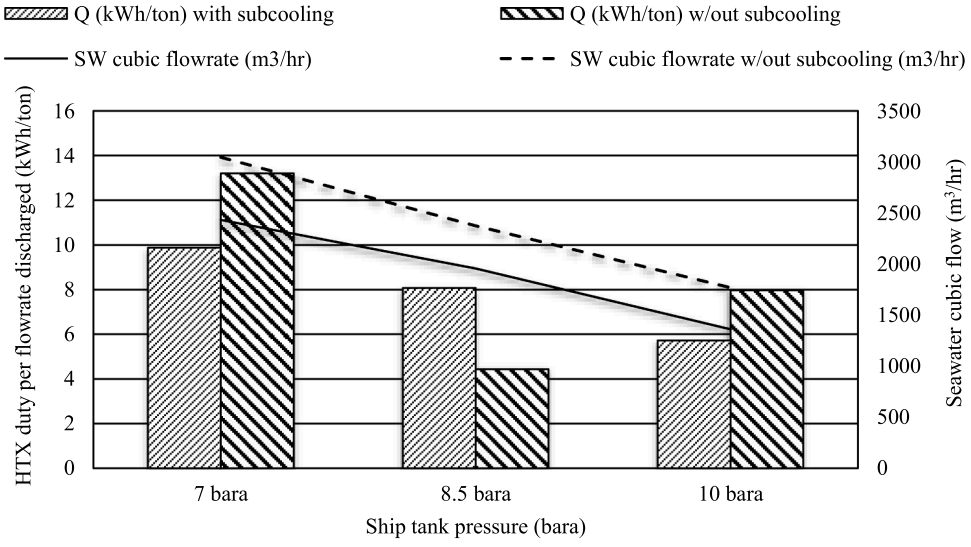


Fig. 10. HTX thermal duty for MP terminal pressure at 16bara.

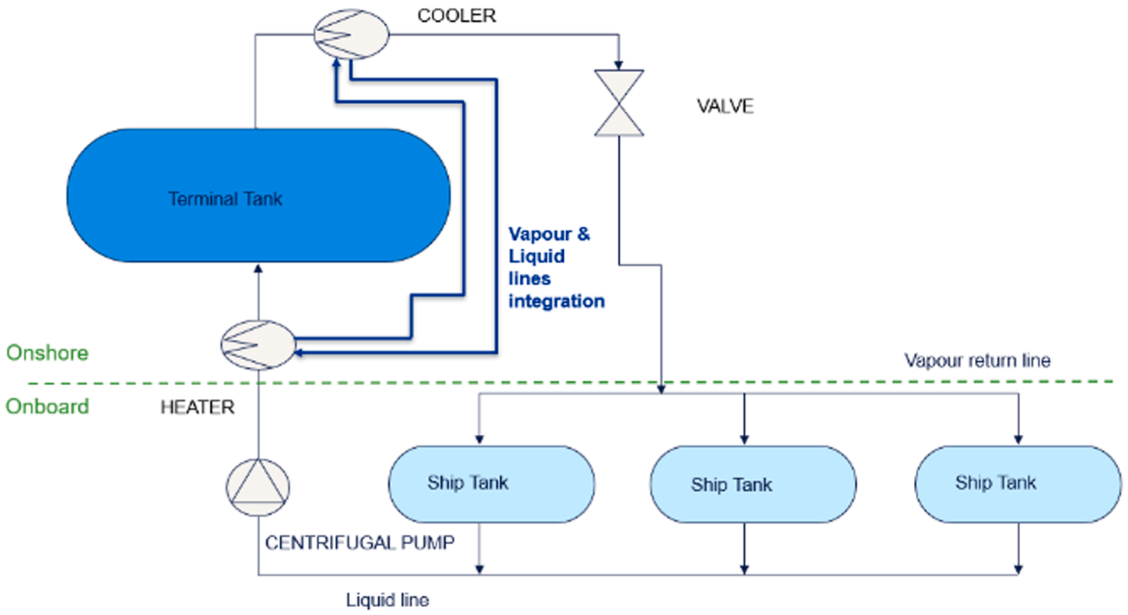


Fig. 11. Process diagram for option 2: Vapor and liquid line integration.

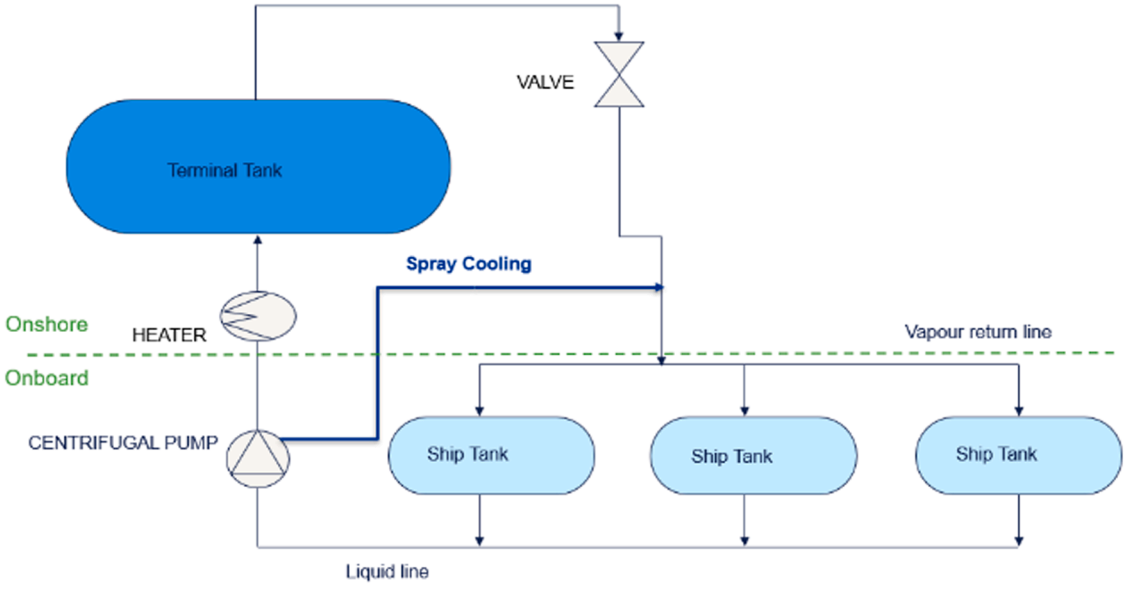


Fig. 12. Process diagram for option 2: Spray Cooling Layout.

$$Q = U * A * LMTD \tag{2}$$

$$LMTD = \frac{\Delta T_A - \Delta T_B}{ln\left(\frac{\Delta T_A}{\Delta T_B}\right)} = \frac{\left(T_{SW_{in}} - T_{CO2_{out}}\right) - \left(T_{SW_{out}} - T_{CO2_{in}}\right)}{ln\left(\frac{\left(T_{SW_{in}} - T_{CO2_{out}}\right)}{\left(T_{SW_{out}} - T_{CO2_{in}}\right)}\right)} \tag{3}$$

Where the below assumptions were made:

- Q, heater thermal duty power in W
- U=2000 W/m^2. °C
- A the estimated HTX surface in m^2
- *LMTD* the logarithmic mean temperature difference, as calculated

For the above design characteristics, the HTX CAPEX was estimated, using Eq. (4), Dimopoulos et al. (2014):

$$HTX\ CAPEX = a * A^b * c^{Pressure} \quad (EUR) \tag{4}$$

where coefficients *a, b* and *c* are empirical and equal 12,003, 0.603, and 1.011187, respectively. The estimated HTX surface (in m^2) is denoted by *A*. The results are shown in the Table 4 per LP-MP combination:

It is noted that these costs are presented to indicate the order of magnitude of equipment cost. Additional costs for auxiliary components and equipment are estimated as a percentage of the core component cost.

Next, the electricity demands for pumping purposes are estimated. For all equipment, a fixed electrical loss of 4 % is assumed along the circuit from the pump motor (duty point) to the switchboard. This loss factor is incorporated into the calculation of total electrical demand to ensure a more accurate representation of system efficiency.

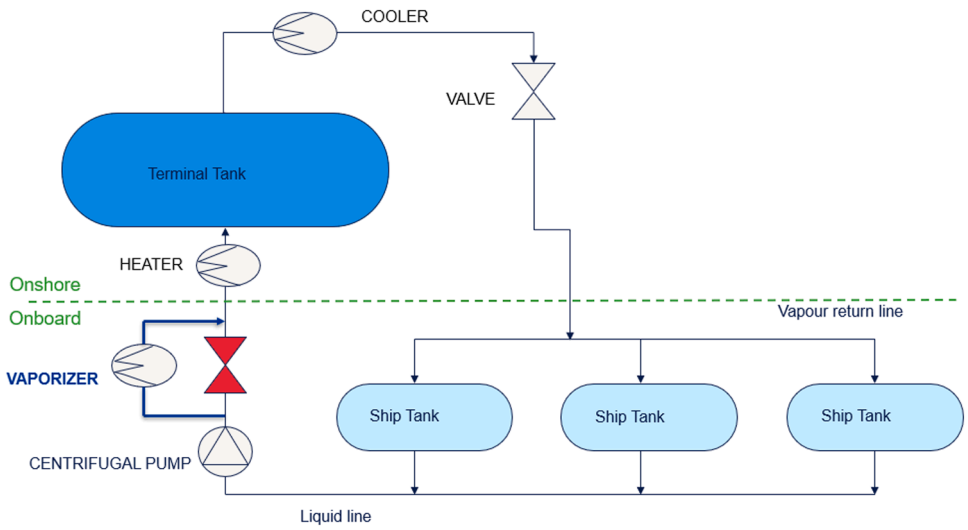


Fig. 13. Discharging process diagram including vaporizer.

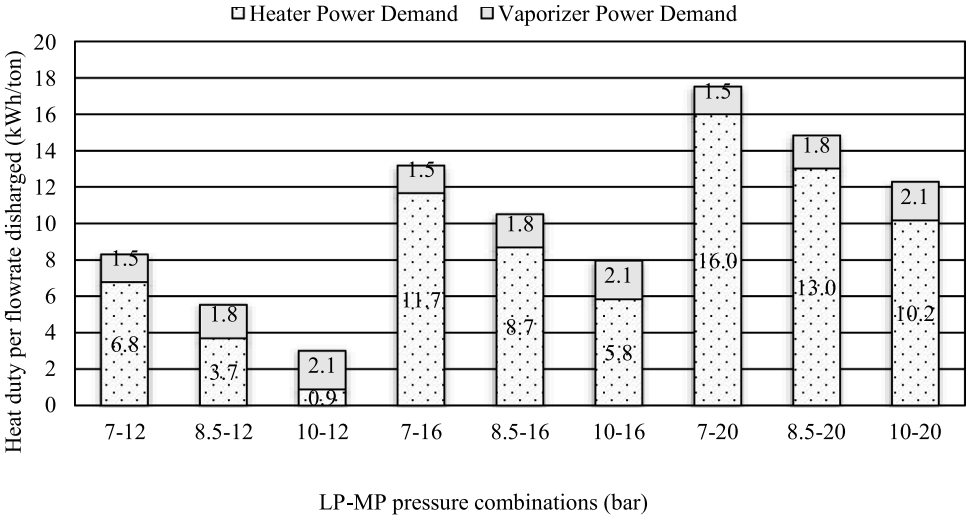


Fig. 14. Total heat demand of vaporizer and HTX.

Table 3
Heater specifications.

Feature	Value	Unit
Capacity	up to 4000	m^3/h
Surface Area	370–2500	m^2

Table 4
Heater CAPEX results.

MP (bara)	LP (bara)	Qs (MW)	LMTD (°C)	Area (m^2)	Cost (EUR)
12	7	15.9	63.1	126	253,842
	8.5	10.4	60.8	86	200,742
	10	5.5	58.7	47	140,157
16	7	25.4	58.5	217	367,556
	8.5	19.8	56.3	176	323,745
	10	14.7	54.4	136	276,852
20	7	33.7	54.7	308	475,104
	8.5	27.9	52.6	266	434,523
	10	22.7	50.7	224	392,118

- Heater liquid line - Seawater flow pumping: The effect of maintaining liquid to subcooling state is beneficial in optimizing both CAPEX and OPEX. This heating is assumed to be provided by HTX on the land side. Based on the seawater flowrate and the thermal duty, the pumping power demand (electrical) is estimated, assuming an acceptable pressure drop of the seawater flow along the HTX (e.g. 2 to 5 bara). This heating duty is provided from shore side:
- Liquid line - Ship Pump operation: This pumping duty is provided from ship side.
- Electrical demand for cooling of vapor line: The thermal demand of the vapor cooler (duty) is converted to electrical through the assumption of a COP efficiency equal to 2.4, Alabdulkarem and Hwang (2011). This is provided from shore side.

The sum of the calculated electric power demand (pump and cooler) is presented in Fig. 15, with a breakdown per ship and shore is given in Fig. 16. Considering all above, and assuming the cost of electric power in Greece to be 0.11 EUR/kWh, the below OPEX results were estimated, as shown in Table 5.

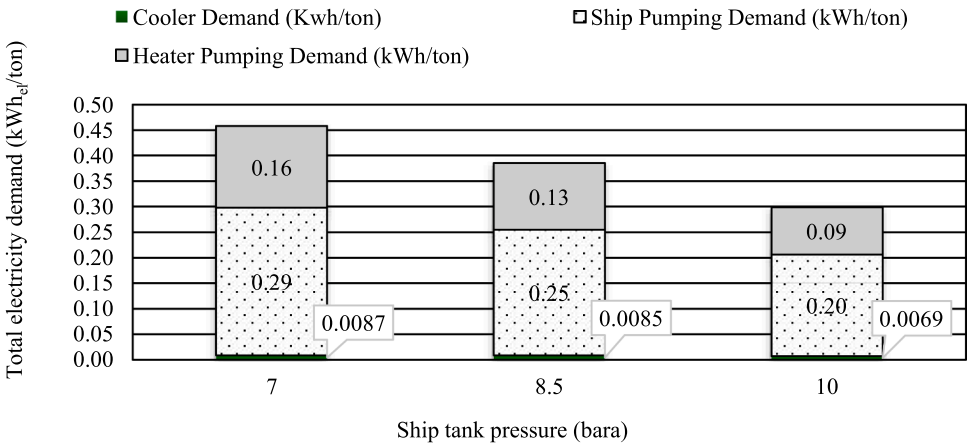


Fig. 15. Breakdown of total electric power demands for LP-MP discharge: Electric demands to cool the liquid line, ship pump power demand, cooler electrical demands.

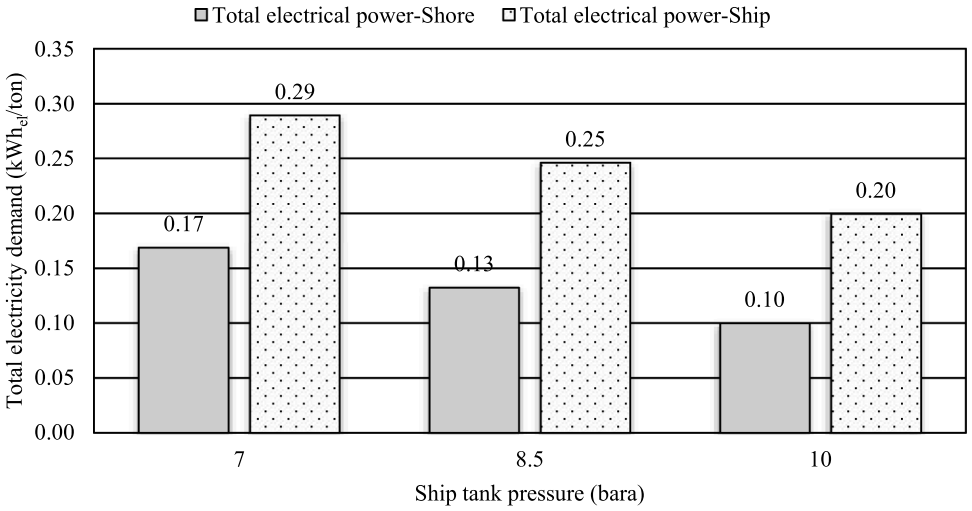


Fig. 16. Total electrical demands, separated by system: Ship and terminal.

Table 5
OPEX for discharging operation for MP at 16bara.

LP (bara)	7	8.5	10
Cost at ship-side (EUR/ton)	0.03	0.03	0.02
Cost at terminal-side (EUR/ton)	0.02	0.01	0.01
LCO_2 mass discharged (tons)	22,970	22,602	22,180

4. MP-LP loading operation

4.1. Evaluation of the MP to LP loading requirements

During loading operations from MP to LP, vapor is displaced from the ship tank due to the pressure differential. A critical aspect of this process is managing vaporization within the loading line, as excessive vapor formation reduces the effective volume of liquid cargo transferred. Minimizing this vaporization is essential to maximize the amount of cargo loaded into the ship tank. The displaced vapor from the ship is returned to the terminal via the VR line, while additional vapor is introduced into the terminal tank at saturation temperature to maintain thermodynamic equilibrium. A sub-cooler of the liquid line is also considered and evaluated as part of the concept, aiming to achieve no vapour formation after the JT valve. The process evaluation is performed with and without the operation of the sub-cooler. In this paragraph, the process is evaluated without sub-cooling.

To analyse this process, simulations are conducted using three different pressure settings for the ship tanks and three for the terminal tanks (see Table 1 for pressure range). Fig. 17 presents the total displaced volume at the end of the loading process. The results indicate that the displaced vapor volume increases with higher ship tank pressures. These findings are presented across a range of ship and terminal pressure combinations, highlighting the relationship between tank pressure and vapor handling requirements during loading.

The total electrical power demand associated with the handling of the displaced VR is also calculated (Fig. 18), incorporating all relevant components. This calculation considers the specified temperature differentials that influence system performance. Fig. 19 presents a sensitivity analysis on the effect of tank volume on the power requirements without sub-cooling, versus the displaced vapour. It is noted that the VR is used for (a) pressure control of the buffer tank, and (b) is further treated by the terminal. Regarding the latter, the need for reliquefaction is considered in the results, in separate, to allow understanding of the process performance (Fig. 20).

Although the focus is on VR, the analysis includes auxiliary systems such as the sub-cooler and VR cooler, which are essential for maintaining operational efficiency. The thermal duties of both the sub-cooler and VR cooler are converted into electrical power requirements using a Coefficient of Performance (COP) of 2.4, as referenced in Alabdulkarem and Hwang (2011). To ensure accuracy in estimating the actual electrical load, a 4 % power loss is assumed across the entire electrical

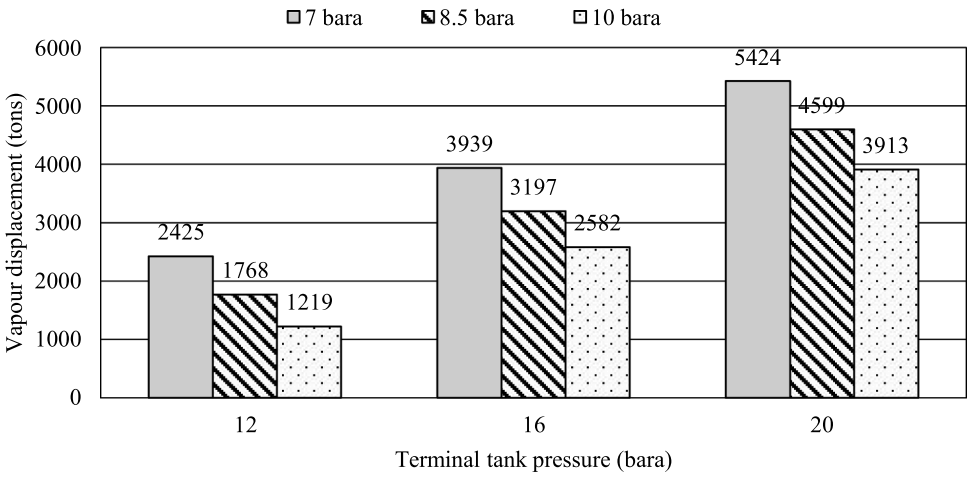


Fig. 17. Total displaced vapor from ship tanks (for the whole loading process) without sub-cooling of the liquid line. Sensitivity over ship tank pressure (7; 8.5; 10 bara).

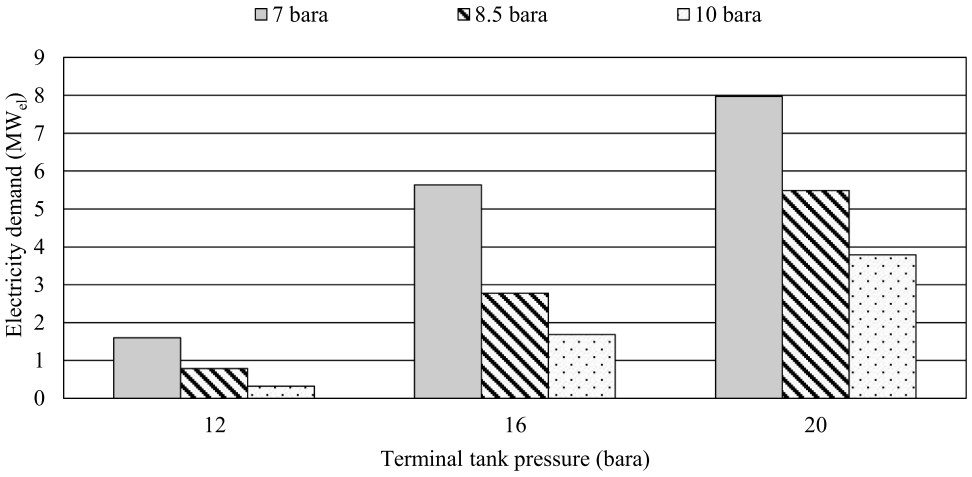


Fig. 18. Electricity demand (VR compressor, cooler), without considering excess VR reliquefaction.

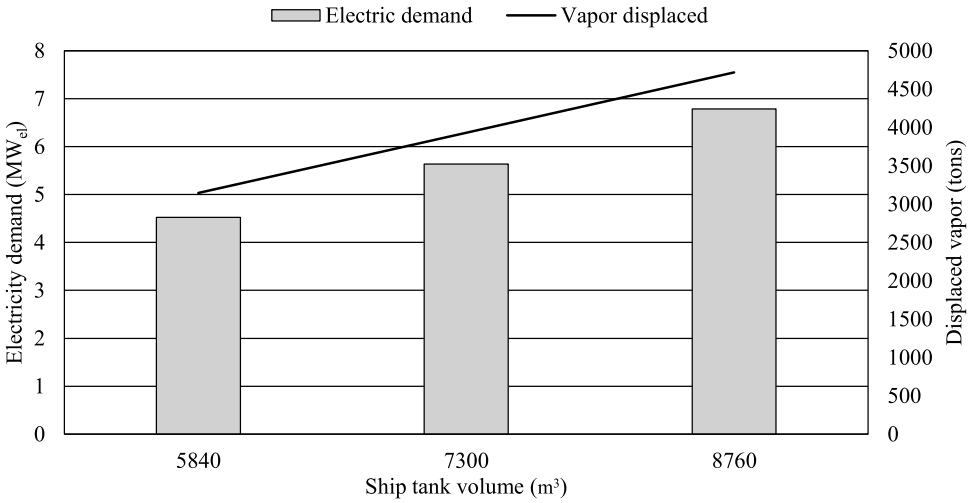


Fig. 19. Sensitivity of electricity demand (VR compressor, cooler) with tank volume, without considering excess VR reliquefaction. Three tanks per ship of the indicated volume are considered. MP is 16bara, LP is 7bara.

distribution circuit-from the pump motor (or other equipment duty points) to the main switchboard. All electrical power required for these systems is supplied from the shore side.

Table 6 shows the results of the two options of VR handling, with and without reliquefaction of the excess vapour produced in the ship tank, and is not used for pressure control of the terminal tank (through VR).

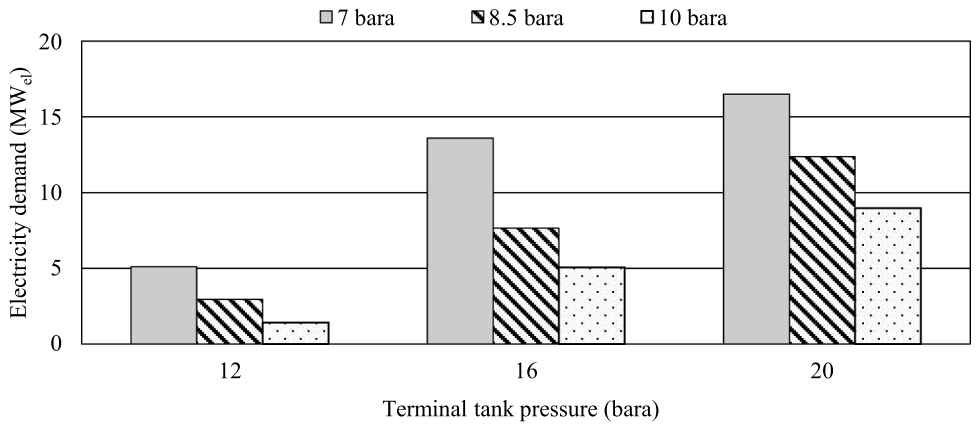


Fig. 20. Total electricity demand including the reliquefaction of the portion of the VR that is not used for pressure maintenance of the terminal tank.

Table 6
Electricity demand breakdown, without sub-cooling. The impact of VR excess reliquefaction is shown.

Low Pressure (bara)	7			8.5			10		
Medium Pressure (bara)	12	16	20	12	16	20	12	16	20
Vapor mass displaced (tons)	4051	6741	9303	2784	5317	7732	1732	4137	6427
Liquid mass loaded (tons)	26,563	26,077	28,746	25,584	26,087	27,364	24,917	25,059	26,199
VR compressor electrical demand (kW_{el})	1134	4090	5792	531	1990	3964	197	1191	2711
VR cooler electrical demand (kW_{el})	466	1548	2177	251	783	1529	123	496	1077
Total electrical demand (kW_{el})	1600	5637	7969	782	2773	5492	319	1687	3788
Total specific electrical demand (kW_{el}/ton)	0.7	2.6	3.3	0.4	1.3	2.4	0.2	0.8	1.7
Operational cost per loaded mass (EUR/tons)	0.08	0.29	0.37	0.04	0.14	0.26	0.02	0.09	0.19
Total loading operational cost (EUR)	2112	7441	10,519	1033	3660	7250	422	2226	5000
VR excess reliquefaction system duty (kW)	3496	7966	8538	2174	4886	6889	1091	3366	5196
Total power including VR excess reliquefaction (kW)	5096	13,603	16,507	2956	7659	12,381	1411	5052	8983
Total specific electrical demand with reliquefaction (kW_{el}/ton)	2.3	6.3	6.9	1.4	3.5	5.4	0.7	2.4	4.1
Total operational cost per loading with reliquefaction (EUR/tons)	0.25	0.69	0.76	0.15	0.39	0.60	0.07	0.27	0.45
Total cost per loading (incl. reliq)	6727	17,956	21,790	3902	10,110	16,343	1862	6669	11,858

The vapor mass displaced is partly used for the maintenance of the terminal pressure and, partly, treated by the terminal. If this excess VR is liquefied, the reliquefaction demand impacts the total power, as compared to the case of the system running without reliquefaction. The simulation is performed for 7, 8.5 and 10 bara (LP ship tank) and 12, 16 and 20 bara (MP terminal tank), respectively.

4.2. Impact of the use of sub-cooling the LCO2 stream

The use of a sub-cooler for the purpose of reducing the temperature of the cargo to a temperature closer to the LP saturation value is evaluated with respect to reduced excess vapor generation, beyond the VR needed for terminal tank pressure maintenance. The simulation was performed for a range of pressures of the LP ship tank (7, 8.5 and 10 bara) and the MP terminal tank (12, 16 and 20 bara), respectively. The subcooling (ΔT) level is defined as the differential between the MP saturated temperature and the sub-cooled temperature to minimize excess VR (Fig. 21).

Table 7 shows the results of VR handling with subcooling where it becomes apparent that the subcooling system requires excessive amount

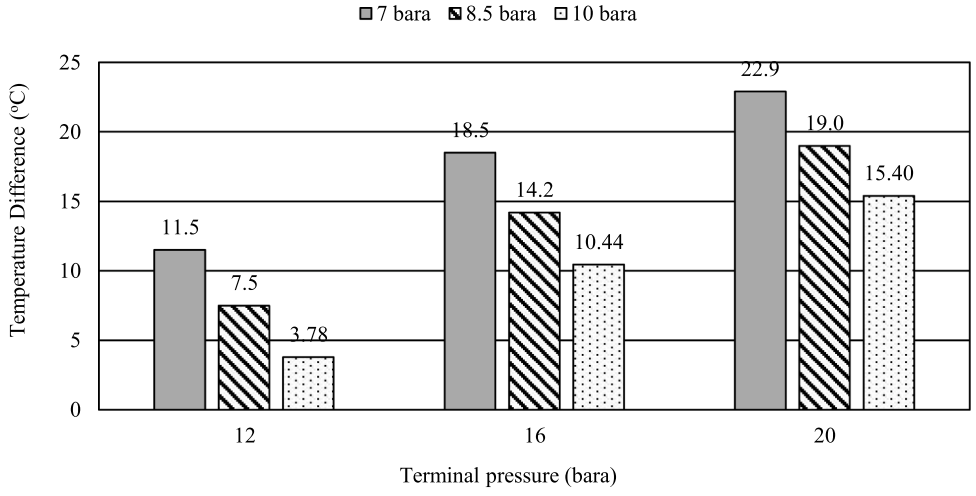


Fig. 21. Required subcooling temperature difference for the elimination of excess vapor generation.

Table 7
Electrical demand breakdown , considering the effect of subcooling.

Low Pressure (bara)	7			8.5			10		
Medium Pressure (bara)	12	16	20	12	16	20	12	16	20
Vapor displaced (tons)	203	271	360	190	259	346	190	260	343
Liquid loaded (tons)	24,190	24,919	25,860	24,425	23,396	23,620	23,910	23,057	23,304
VR compressor electrical demand (kW_{el})	405	1083	1910	306	730	1105	111	452	904
VR cooler electrical demand (kW_{el})	62	0	1	0	−47	−18	34	23	20
Reliquefaction system duty (kW_{el})	204	449	736	167	326	459	83	211	385
Sub-cooler duty (kW)	5344	8964	12,046	3558	6644	9245	1774	4872	7486
Total duty with subcooling (kW)	6015	10,495	14,693	4031	7653	10,790	2001	5557	8795
Total specific electrical demand with subcooling (kW_{el}/ton)	3.0	5.1	6.8	2.0	3.9	5.5	1.0	2.9	4.5
Operational cost per loaded mass with subcooling (EUR/ton)	0.33	0.56	0.75	0.22	0.43	0.60	0.11	0.32	0.50
Total operational cost per loading with subcooling (EUR)	7940	13,854	19,394	5321	10,102	14,243	2642	7336	11,609

of power as compared to the power demand of the system running without it. However, in comparison with the total power demand of the process, including the reliquefaction system demands, the sub-cooler is less costly. The simulation is performed for 7, 8.5 and 10 bara (LP ship tank) and 12, 16 and 20 bara (MP terminal tank), respectively. (Fig. 22)

5. Results and conclusions

The study investigated the compatibility between low-pressure (LP, 7–10 bara) CO_2 carriers and medium-pressure (MP, 12–20 bara) terminals, a key challenge in scaling carbon capture and storage (CCS) infrastructure. The study demonstrated that the LP-MP discharging interfacing is feasible without requiring major additions to the equipment list on either side, while the MP-LP loading interfacing is feasible under decisions for the handling of the vapour return. Using DNV's in-house COSSMOS simulation framework, the thermodynamic and operational interfacing between a LCO_2 carrier (three tanks ranging from 5840 to 8760 m^3) and a 30,000 m^3 MP terminal was analysed.

Through simulation of discharging and loading operations, the study showed that the pressure mismatch can be managed, with operational penalties ranging within 0.03 to 0.05 EUR/ton (per ton discharged), and 0.07 to 0.76 EUR/ton (per ton loaded), respectively. The study identified that LP-to-MP discharging can be optimized through subcooling, which reduces vapor displacement and electricity demand. Conversely, MP-to-LP loading requires either vapor return compression and cooling, or subcooling to minimize VR. The total energy penalty, accounting for all process requirements- whether involving vapor return reliquefaction or liquid stream subcooling -falls within comparable ranges, with the option of no sub-cooling demonstrating slightly lower cost for LP equal and above 8.5bara. In summary, the study demonstrated that:

- Vapor return during discharging is naturally driven by the ship tank pressure, simplifying operations.
- Subcooling the LCO_2 stream during LP-to-MP transfer promotes condensation in the terminal tank, reducing vapor displacement and thermal load.
- Integration of "cold" energy from the LP stream into the MP system is energetically sound, though infrastructure complexity must be considered.
- Vapor management during loading is achievable without additional equipment, at the expense of energy penalty (0.07 to 0.76 EUR/ton). Sub-coolers demonstrate slightly higher electricity demand (0.11 to 0.75 EUR/ton).

The study concludes that with targeted strategies, such as subcooling and vapor management, LP-MP CO_2 transfer can be achieved efficiently, supporting flexible CCS logistics.

CRediT authorship contribution statement

Panos Deligiannis: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Vangelis Karageorgos:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **Paraskevi Bardavillia:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Chara Georgopoulou:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal

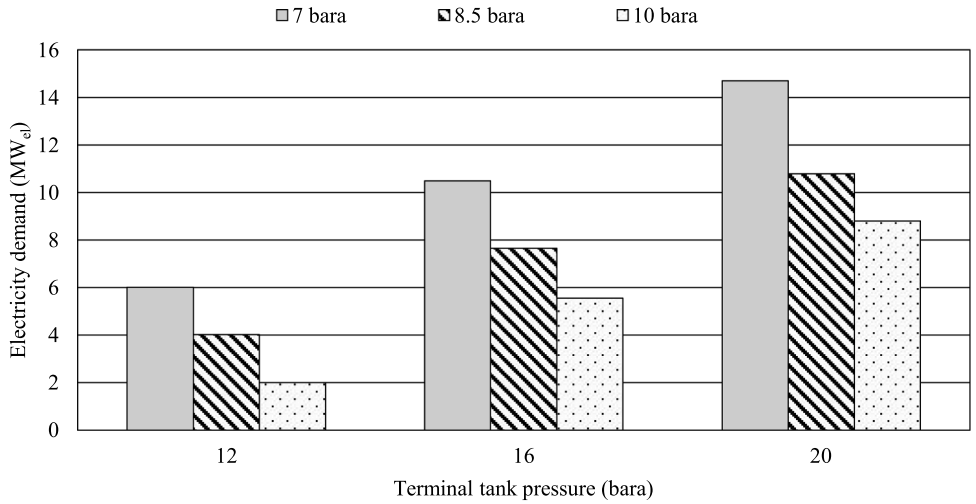


Fig. 22. Total electrical demand considering the effect of subcooling.

relationships which may be considered as potential competing interests: Chara Georgopoulou reports a relationship with DNV Hellas Single Member S.A. that includes: employment. Panos Deligiannis reports a relationship with Ecolog Ltd. that includes: employment. Vangelis Karageorgos reports a relationship with Ecolog Ltd. that includes: employment. Paraskevi Bardavillia reports a relationship with DNV Hellas Single Member S.A. that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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